Every Algebraic Number is a Critical Value of an Integer Polynomial

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Abstract

For every nonconstant $f \in \mathbb{Z}[X]$ we construct $g \in \mathbb{Z}[X]$ and an irreducible $H \in \mathbb{Z}[X]$ with $H \mid g'$ and $H^2 \mid f \circ g$. Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of $\mathbb{Z}[X]$ is exactly $\overline{\mathbb{Q}}$. The construction is explicit. The proof was assisted by AI, and has been formalized in Lean.

1 Introduction

A critical value of a nonconstant $g \in \mathbb{Z}[X]$ is a number α with $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some β. Since $\overline{\mathbb{Q}}$ is closed under polynomial evaluation, every critical value is algebraic. This note proves the converse, and does so with an explicit construction.

Theorem 1. *Let $f \in \mathbb{Z}[X]$ be nonconstant. Then there exist $g, H \in \mathbb{Z}[X]$ with H irreducible in $\mathbb{Z}[X]$ and $1 \leq \deg H$, such that*

$$H \mid g' \quad \textit{and} \quad H^2 \mid f \circ g \qquad \textit{in } \mathbb{Z}[X].$$

The g produced satisfies $\deg g \geq 2$.

The corollary is the statement in the title.

Corollary 1. *The set of critical values is exactly $\overline{\mathbb{Q}}$.*

1.1 Conventions

Throughout, $p' =$ derivative p, and for a fixed nonzero $M \in \mathbb{Z}$ we write

$$\tilde{p} := p(MX) = p \circ (MX),$$

the *dilation* of p. Dilation is a ring homomorphism $\mathbb{Z}[X] \to \mathbb{Z}[X]$, it is injective for $M \neq 0$, it preserves degree, and it interacts with the derivative by the chain rule

$$(\tilde{p})' = M\,\widetilde{p'}. \tag{1}$$

These four facts are used on essentially every line of §3 and are worth isolating before anything else; the factor M in (1) is the reason M has to appear in the statement of Lemma 2.

2 The construction

Fix a nonconstant $f \in \mathbb{Z}[X]$ and suppose $X \nmid f$; the excluded case is settled separately in §6. All of the following data is explicit and computable.

Symbol	*Definition*	*Notes*
h	a primitive irreducible factor of f with $h(0) \neq 0$	in $\mathbb{Z}[X]$
f_1	the cofactor, $f = hf_1$	in $\mathbb{Z}[X]$
n	$\deg h \geq 1$	
h_0	$h(0) \neq 0$	in $\mathbb{Z}$
u, v, ρ	any Bézout data, $uh + vh' = \rho$, $\rho \neq 0$	$u, v \in \mathbb{Z}[X]$, $\rho \in \mathbb{Z}$
M	$\rho\, h_0$, so $M \neq 0$	the dilation factor
H	$h(MX)/h_0$	$H(0) = 1$, $\deg H = n$
W	$(v(0)H - v(MX))/\rho$	$W(0) = 0$
g	$MX + h(MX)\,W \ = \ MX + h_0 HW$	

Existence of h. f is nonconstant, so it has an irreducible factor. Irreducible factors of positive degree in $\mathbb{Z}[X]$ are primitive, and $h(0) \neq 0$ because $h(0) = 0$ would give $X \mid h \mid f$, excluded.

Existence of the Bézout data. h is irreducible and primitive in $\mathbb{Z}[X]$, hence irreducible in $\mathbb{Q}[X]$ by Gauss's lemma. In characteristic zero $h' \neq 0$ and $\deg h' < \deg h$, so $h \nmid h'$ and therefore $\gcd(h, h') = 1$ in the principal ideal domain $\mathbb{Q}[X]$. Bézout there and clearing denominators produces $u, v \in \mathbb{Z}[X]$ and $\rho \in \mathbb{Z} \setminus \{0\}$.

Any nonzero ρ in the elimination ideal $(h, h') \cap \mathbb{Z}$ works. The resultant $\mathrm{Res}(h, h')$ is one admissible choice and is never needed as such; smaller choices give a smaller g, and the generator of the elimination ideal — the *reduced* resultant — is the best available. Example 2 below uses $\rho = 2$ where the resultant is 4.

2.1 Integrality

The two divisions in the table are exact, which is the whole reason M was defined as the product ρh_0 rather than as either factor alone.

Lemma 1 (Integrality). *$H \in \mathbb{Z}[X]$ with $H(0) = 1$ and $\deg H = n$; and $W \in \mathbb{Z}[X]$ with $W(0) = 0$.*

Proof. Write $h = \sum_j h_j X^j$ and $v = \sum_j v_j X^j$. The coefficients of H are $H_0 = h_0/h_0 = 1$ and, for $j \geq 1$,

$$H_j \ = \ \frac{h_j M^j}{h_0} \ = \ \frac{h_j\, \rho^j h_0^j}{h_0} \ = \ h_j\, \rho^j h_0^{j-1} \ \in \ \mathbb{Z},$$

using $h_0 \mid M$. Since $\deg h = n$ and $M \neq 0$, $\deg H = n$.

For W, the coefficient of X^j in $\rho W = v(0)H - \tilde{v}$ is $v(0)H_j - v_j M^j$. At $j = 0$ this is $v_0 \cdot 1 - v_0 = 0$, which gives both integrality there and $W(0) = 0$. For $j \geq 1$ the display above exhibits $\rho^j \mid H_j$, and $\rho \mid M^j$, so ρ divides the whole coefficient. □

Remark 1. For the proof of Theorem 1 it is cleaner to treat H and W as *data* subject to the defining identities

$$h_0 H = \tilde{h}, \qquad \rho W = v(0)H - \tilde{v}, \qquad H(0) = 1, \qquad W(0) = 0, \tag{2}$$

rather than as quotients. Nothing after this point uses anything else about them, and no division in $\mathbb{Z}[X]$ ever has to be justified again. Lemma 1 is then exactly the statement that such data exists.

3 The two identities

Both statements below are exact identities in $\mathbb{Z}[X]$. There are no congruences, no roots and no primality anywhere in this section; in particular *irreducibility of h is never used here*. It is needed only to produce the Bézout data in §2 and again in §5.2.

Dilating the Bézout relation $uh + vh' = \rho$ by the ring homomorphism $p \mapsto \tilde{p}$ gives the form in which it will be used,

$$\tilde{u}\,\tilde{h} + \tilde{v}\,\widetilde{h'} = \rho, \qquad \text{equivalently} \qquad \tilde{v}\,\widetilde{h'} = \rho - h_0\tilde{u}H, \tag{3}$$

the second form by $\tilde{h} = h_0 H$.

Lemma 2 (Derivative). *$\rho\, g' = HA$, where $A := Mh_0\tilde{u} + Mv(0)\widetilde{h'} + \rho h_0 W' \in \mathbb{Z}[X]$.*

Proof. From $g = MX + \tilde{h}W$ and the chain rule (1),

$$g' \;=\; M + M\widetilde{h'}W + \tilde{h}\,W'.$$

Multiply by ρ and substitute $\rho W = v(0)H - \tilde{v}$ and $\tilde{h} = h_0 H$:

$$\rho g' \;=\; \rho M + M\widetilde{h'}(v(0)H - \tilde{v}) + h_0 H\rho W' \;=\; \rho M - M\widetilde{h'}\,\tilde{v} + H(Mv(0)\widetilde{h'} + \rho h_0 W').$$

By (3), $\rho M - M\widetilde{h'}\tilde{v} = \rho M - M(\rho - h_0\tilde{u}H) = Mh_0\tilde{u}H$, and collecting the factor H gives the claim. $\square$

Lemma 3 (Value). *$\rho\,(f \circ g) = H^2 D$ for an explicit $D \in \mathbb{Z}[X]$.*

Proof. Put $B := h_0 HW$, so that $g = MX + B$. By Taylor to second order (Lemma 4) there is $k \in \mathbb{Z}[X]$ with

$$f \circ g \;=\; \tilde{f} + \widetilde{f'}B + B^2 k.$$

Since $f = hf_1$ we have $\tilde{f} = \tilde{h}\,\widetilde{f_1} = h_0 H\widetilde{f_1}$, and $B^2 = h_0^2 H^2 W^2$, so

$$f \circ g \;=\; h_0 H(\widetilde{f_1} + \widetilde{f'}W) \;+\; H^2(h_0^2 W^2 k). \tag{4}$$

It remains to extract one more factor of H from the first bracket. Dilating the product rule $f' = h' f_1 + hf_1'$ and multiplying by $\tilde{v}$,

$$\widetilde{f'}\,\tilde{v} \;=\; \widetilde{h'}\,\tilde{v}\,\widetilde{f_1} + \tilde{h}\,\widetilde{f_1'}\,\tilde{v} \;=\; (\rho - h_0\tilde{u}H)\widetilde{f_1} + h_0 H\widetilde{f_1'}\,\tilde{v},$$

using (3) again. Hence $\rho\widetilde{f_1} - \widetilde{f'}\tilde{v} = h_0 H(\tilde{u}\,\widetilde{f_1} - \tilde{v}\,\widetilde{f_1'})$, and so

$$\rho(\widetilde{f_1} + \widetilde{f'}W) \;=\; \rho\widetilde{f_1} + \widetilde{f'}(v(0)H - \tilde{v}) \;=\; HC, \qquad C := h_0(\tilde{u}\,\widetilde{f_1} - \tilde{v}\,\widetilde{f_1'}) + v(0)\widetilde{f'}.$$

Multiplying (4) by ρ and substituting gives $\rho(f \circ g) = H^2 D$ with $D := h_0 C + \rho h_0^2 W^2 k$. $\square$

Lemma 4 (Taylor to second order). *Let R be a commutative ring, $f \in \mathbb{Z}[X]$ and $a, b \in R$. Then there is $k \in R$ with $f(a+b) = f(a) + f'(a)\,b + b^2 k$.*

Proof. Both sides are additive in f and the statement is stable under multiplication by a constant, so it suffices to check $f = X^m$, where the binomial theorem gives $(a+b)^m = a^m + ma^{m-1}b + b^2\sum_{i\geq 2}\binom{m}{i}a^{m-i}b^{i-2}$. Applied with $R = \mathbb{Z}[X]$, $a = MX$ and $b = B$, this is the form used in Lemma 3. $\square$

4 Descent: removing ρ

Lemmas 2 and 3 give the desired divisibilities only after multiplication by ρ. Removing it is where the normalization $H(0) = 1$ earns its place.

Lemma 5 (Descent). *Let $P, K \in \mathbb{Z}[X]$ and $\rho \in \mathbb{Z}$, $\rho \neq 0$, and suppose $K(0)$ is a unit of $\mathbb{Z}$. If $K \mid \rho P$ in $\mathbb{Z}[X]$ then $K \mid P$ in $\mathbb{Z}[X]$.*

Proof. By strong induction on the number of prime factors of $|\rho|$. If $\rho = \pm 1$ there is nothing to prove. Otherwise choose a prime $p \mid \rho$ and write $\rho = p\rho_1$, and let $\rho P = KD$. Reducing modulo p gives $0 = \overline{K}\,\overline{D}$ in $\mathbb{F}_p[X]$. Now $K(0) = \pm 1 \not\equiv 0 \pmod p$, so $\overline{K} \neq 0$; and $\mathbb{F}_p[X]$ is an integral domain, so $\overline{D} = 0$, that is $D = pD_1$. Cancelling p in the domain $\mathbb{Z}[X]$ leaves $\rho_1 P = KD_1$, and $|\rho_1|$ has one prime factor fewer. □

Corollary 2. *$H \mid g'$ and $H^2 \mid f \circ g$, both in $\mathbb{Z}[X]$.*

Proof. Apply Lemma 5 to Lemma 2 with $K = H$, whose constant coefficient is 1; and to Lemma 3 with $K = H^2$, whose constant coefficient is likewise 1. □

Remark 2. This is why the construction normalizes $H(0) = 1$ instead of making H monic. Two other routes reach Corollary 2. One: $H(0) = 1$ makes H a unit in the power series ring $\mathbb{Z}[[X]]$, so $(f \circ g)H^{-2}$ lies in $\mathbb{Z}[[X]]$; it also equals $D/\rho \in \mathbb{Q}[X]$; and $\mathbb{Q}[X] \cap \mathbb{Z}[[X]] = \mathbb{Z}[X]$. Two: H is primitive and irreducible, hence prime in $\mathbb{Z}[X]$, and $H \nmid \rho$ by degree. The second route needs §5.2 first, while the induction above and the power series argument need nothing.

5 Nondegeneracy and irreducibility

5.1 Nondegeneracy

Lemma 6. *$W \neq 0$, $\deg W \geq 1$, $\deg g = n + \deg W \geq 2$, and $g' \neq 0$.*

Proof. Suppose $W = 0$. Then $\tilde{v} = v(0)H$, and comparing with $h_0 H = \tilde{h}$ gives $h_0\tilde{v} = v(0)\,\tilde{h}$; since dilation is injective, $h_0 v = v(0)h$. Then $h \mid h_0 v$ in $\mathbb{Q}[X]$, so $v = (v(0)/h_0)h$, and the Bézout relation collapses to $\rho = h(u + (v(0)/h_0)h')$, forcing $\deg h = 0$ and contradicting $n \geq 1$. So $W \neq 0$, and $W(0) = 0$ then gives $\deg W \geq 1$.

Since $M \neq 0$ we have $\deg \tilde{h} = n$, so $\deg(\tilde{h}W) = n + \deg W \geq n+1 \geq 2 > 1 = \deg(MX)$. There is therefore no cancellation in $g = MX + \tilde{h}W$ and $\deg g = n + \deg W$. Finally $\deg g \geq 2 \geq 1$ and characteristic zero give $g' \neq 0$. □

If the Bézout data is chosen with $\deg v < n$, which is automatic for the subresultant construction, then $\deg W \leq n$ and hence

$$n + 1 \;\leq\; \deg g \;\leq\; 2n.$$

The lower bound is not an artifact: $H \mid g'$ with $\deg H = n$ forces $\deg g' \geq n$ for *any* solution whatsoever, so no construction can do better. Example 2 attains it.

5.2 Irreducibility

Lemma 7. *H is irreducible in $\mathbb{Z}[X]$.*

Proof. h is primitive and irreducible in $\mathbb{Z}[X]$, hence irreducible in $\mathbb{Q}[X]$ by Gauss's lemma. For $M \neq 0$ the dilation $X \mapsto MX$ is a $\mathbb{Q}$-algebra automorphism of $\mathbb{Q}[X]$, so $\tilde{h}$ is irreducible in $\mathbb{Q}[X]$; and $H = \tilde{h}/h_0$ differs from it by a unit of $\mathbb{Q}$, so H is irreducible in $\mathbb{Q}[X]$. By Lemma 1, $H(0) = 1$, so H is primitive. Primitive and irreducible over $\mathbb{Q}$ gives irreducible in $\mathbb{Z}[X]$, by Gauss's lemma again. □

This is the only place Gauss's lemma is used, and $\mathbb{Z}$ enters the argument only here and in the prime factorization of Lemma 5. Everything else goes through over any characteristic zero unique factorization domain.

6 The degenerate case

Suppose $X \mid f$. Then

$$g = X^2, \qquad H = X$$

settles Theorem 1: $g' = 2X$ is divisible by H; writing $f = Xf_2$ gives $f \circ g = X^2\,(f_2 \circ X^2)$, which is divisible by $H^2 = X^2$; and X is irreducible in $\mathbb{Z}[X]$ of degree 1, with $\deg g = 2$.

Note that the case split is on $X \mid f$, not on f being a monomial: this branch covers $f = X(X^2+1)$ as well as $f = cX^k$. Equivalently, the split is on whether f has *some* irreducible factor with nonzero constant term.

Proof of Theorem 1. If $X \mid f$, use §6. Otherwise take h, and Bézout data, as in §2, and define M, H, W, g by the table; these lie in $\mathbb{Z}[X]$ by Lemma 1. Then H is irreducible by Lemma 7 with $\deg H = n \geq 1$, the two divisibilities hold by Corollary 2, and $\deg g \geq 2$ by Lemma 6. $\square$

7 Critical values

Only now do roots appear.

Proof of Corollary 1. Let $\alpha \in \overline{\mathbb{Q}}$. If $\alpha = 0$, take $g = X^2$ and $\beta = 0$. Otherwise let h be the primitive integral minimal polynomial of α, so $h(0) \neq 0$, and run §2 with $f = h$, hence $f_1 = 1$. Set

$$\beta := \alpha/M.$$

Then $H(\beta) = h(M\beta)/h_0 = h(\alpha)/h_0 = 0$. Evaluating the Bézout relation at α and using $h(\alpha) = 0$ gives $v(\alpha)h'(\alpha) = \rho$, so

$$W(\beta) = \frac{v(0)H(\beta) - v(\alpha)}{\rho} = -\frac{v(\alpha)}{\rho} = -\frac{1}{h'(\alpha)},$$

which is defined because h is separable. Therefore

$$g(\beta) = M\beta + h(\alpha)W(\beta) = \alpha, \qquad g'(\beta) = M + Mh'(\alpha)W(\beta) + h(\alpha)W'(\beta) = M(1-1) = 0.$$

So β is a critical point of g with critical value α. The converse was observed at the start of §1. $\square$

Two features of this are worth naming. First, β is the *same* rescaling α/M for every root α of h simultaneously: a single g has all n conjugates as critical values, at the n points α_i/M. Second, the ring at work is $\mathbb{Z}[\beta] = \mathbb{Z}[\alpha/M] \supseteq \mathbb{Z}[\alpha][1/\rho]$. Dividing α by M is what makes the higher Taylor coefficients integral, while $H(0) = 1$ makes β a *unit* in $\mathbb{Z}[\beta]$, and that is what permits trading the constant term of $1/h'(\alpha)$ for higher degree terms.

What does *not* appear is as much to the point: no class group of $\mathbb{Q}(\alpha)$, no unit group $\mathcal{O}_K^\times$, no monogenicity of $\mathcal{O}_K$, and no Thue equation or index form. $\mathbb{Z}[\beta]$ is not an order in $\mathcal{O}_K$ — β is not an algebraic integer — and the units the argument consumes are S-units manufactured by the construction itself.

For a reducible f the statement reduces to the irreducible case for a reason worth stating explicitly, since it is easy to mis-handle: a root α of f is a root of *some* irreducible factor h of f, and the construction run on that h produces a g having α as a critical value. Different factors give different g. It is not the case that one g serves all of a reducible f.

8 Examples

Each line below is a closed identity, checkable by expansion.

Example 1 ($f = qX - p$, $p \neq 0$). With $u = 0$, $v = 1$, $\rho = q$, $h_0 = -p$, $M = -pq$, $H = q^2X + 1$ and $W = qX$,

$$g = -pq^3X^2 - 2pqX, \qquad g' = -2pq\,(q^2X + 1), \qquad f \circ g = -p\,(q^2X + 1)^2.$$

Example 2 ($f = X^2 + 1$). With $u = 2$, $v = -X$, $\rho = 2$, $h_0 = 1$, $M = 2$, $H = 4X^2 + 1$ and $W = X$,

$$g = 4X^3 + 3X, \qquad g' = 3(4X^2 + 1), \qquad f \circ g = (4X^2 + 1)^2(X^2 + 1).$$

Here $\deg g = 3 = n + 1$ attains the lower bound of §5.1. Note also $\rho = 2$ while $\mathrm{Res}(h, h') = 4$: the reduced resultant is the better choice. The unit group $\mathcal{O}_K^\times = \{\pm 1, \pm i\}$ has rank 0, and it is $\mathbb{Z}[i/2] = \mathbb{Z}[i][1/2]$ that supplies the needed unit $1 + i$.

Example 3 ($f = X^3 - 2$). With $u = 3$, $v = -X$, $\rho = -6$, $h_0 = -2$, $M = 12$, $H = 1 - 864X^3$ and $W = -2X$,

$$g = -3456X^4 + 16X, \qquad g' = 16\,(1 - 864X^3).$$

Example 4 ($f = X^3 - X - 1$). The construction gives $M = 23$ and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334X^5 + 39675X^4 - 26450X^3 + 9085X^2 - 1610X + 118, \qquad H = 23X^3 - 23X^2 + 8X - 1,$$

with $g' = -230\,(23X - 7)H$ and $H^2 \mid f \circ g$, the cofactor having degree 9.

Example 5 (Dedekind's cubic). $f = X^3 - X^2 - 2X - 8$ is the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. The construction runs unchanged, which is the point of including it — no step of §2–§4 is sensitive to monogenicity.

It is also where formalization corrected the author. An earlier draft of this material asserted $M = 4024$ here. That is unattainable. Since $M = \rho h_0$ and $h_0 = -8$, it would need $\rho = -503$, the discriminant of the field; but every Bézout constant for this h is even. Writing $h = X^2(X - 1) - 2(X + 4)$ and $h' = 3X^2 - 2(X + 1)$ puts any relation $uh + vh' = c$ into the shape

$$X^2\big(u(X - 1) + 3v\big) \;=\; c + 2\big(u(X + 4) + v(X + 1)\big),$$

and reading off the constant coefficient gives $0 = c + 2\big(u(X + 4) + v(X + 1)\big)(0)$, so $2 \mid c$. In fact $(h, h') \cap \mathbb{Z} = (1006)$, with certificate

$$(125 - 21X)\,h + (3 - 44X + 7X^2)\,h' = -1006,$$

so the smallest attainable $|M|$ is 8048. The relevant invariant is the reduced resultant, not $\mathrm{disc}(K) = -503$; here $\mathrm{disc}(h) = -2012$ and the index is 2.

9 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
    exists g H : Z[X], Irreducible H /\ H | derivative g /\ H^2 | f.comp g
```

with a strengthened form carrying $2 \le \deg g$ and $1 \le \deg H$, and with Corollary 1 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

The organization follows the sections above: the dilation and its chain rule (1) first, then `Setup`, a structure carrying the data of §2 with (2) as hypotheses, then Lemmas 2 and 3 as `linear_combination` calls from those hypotheses, then descent, irreducibility and nondegeneracy. Treating H and W as data rather than as quotients, as in Remark 1, is what keeps §3 free of any reasoning about divisibility in $\mathbb{Z}[X]$; the integrality of Lemma 1 is discharged once, in the existence proof for `Setup`.

Three things were genuinely easier on paper than in Lean, and are recorded in case they are useful to someone attempting a similar development. Dilation as a ring homomorphism, and its interaction with `derivative`, is used everywhere and deserves to be isolated first. Lemma 4 is stated in Mathlib for `eval` rather than for `comp`, and has to be transported. And Mathlib has the translation $X \mapsto X + c$ as an algebra equivalence but not the dilation $X \mapsto cX$, which had to be built to prove Lemma 7.

The one place where the formalization changed the mathematics is Example 5, whose erratum is itself machine-checked.

10 Computation

The construction is effective, and a program implementing it accompanies this note. It takes $f \in \mathbb{Z}[X]$, factors it, and for each irreducible factor h returns M, H, W, g and a verification of every hypothesis and conclusion of §2–§5.2 on the actual output.

Two remarks on what the computation costs. The Bézout data is obtained by the extended Euclidean algorithm in $\mathbb{Q}[X]$ followed by content reduction, which yields the generator of $(h, h') \cap \mathbb{Z}$ — the reduced resultant, and so the smallest $|M|$ available. Factoring f is the expensive step and is delegated to FLINT [Har10], which implements van Hoeij's knapsack recombination [Hoe02] in the improved form of [HNH11]. Root finding, used only for display, is the Aberth–Ehrlich iteration [Abe73], the method underlying MPSolve [BF00].

The coefficients grow quickly, and it is worth being blunt about the size. Since $\deg g = n + \deg W \le 2n$ and the coefficients of H carry $\rho^j h_0^{j-1}$, the output is large even for small input: $f = x^5 - x - 1$ gives a g of degree 10 with 34-digit coefficients, and $f = x^{10} + 20x^4 + x^3 - 10x^2 + 1$ gives degree 20 with 376-digit coefficients. This is not a defect of the implementation; it is the size of the answer.

11 Appendix: provenance of the imported results

A formalization is only as attributable as its imports. This appendix records, for every Mathlib result the development invokes by name, where the mathematics came from.

The list was extracted mechanically rather than by memory: a Lean metaprogram resolves every identifier appearing in the sources against the environment and reports those that Mathlib owns, together with the module defining each. It finds about sixty named results across fifty modules.

The first thing that extraction shows is that Mathlib itself attributes none of them. Of Mathlib's $8{,}264$ files, 862 carry a `## References` section; of the fifty this development depends on, *none* does. Its bibliography runs to 484 entries and contains nothing for Gauss's lemma, Bézout's identity, Taylor's theorem or Dedekind. That is not an oversight. It is a fair description of the material: everything Theorem 1 imports is classical, and almost all of it predates 1900. The modern content of the development is the construction of §2 and the descent of §4, neither of which is imported from anywhere.

Result used	*Mathlib declarations*	*Origin*
Gauss's lemma; content and primitive part	`IsPrimitive.Int.irreducible_iff_irreducible_map_cast`, `primPart_mul`, `isPrimitive_primPart`, `primPart_dvd`, `content_eq_zero_iff`	[Gau01], art. 42
Unique factorization; existence of an irreducible factor	`UniqueFactorizationMonoid.factors`, `factors_prod`, `irreducible_of_factor`, `WfDvdMonoid.exists_irreducible_factor`	[Gau01], art. 16; [Coh68; AAZ90]
Bézout in $\mathbb{Q}[X]$	`Irreducible.coprime_iff_not_dvd`	Euclid VII.1–2; [Béz79]
Clearing denominators (localization)	`IsLocalization.integerNormalization`, `integerNormalization_spec`, `nonZeroDivisors`	[Gre27]
Formal derivative; product and chain rules	`derivative_mul`, `derivative_comp`, `derivative_map`, `natDegree_derivative_lt`, `derivative_ne_zero`	[Lag97]
Taylor to second order (Lemma 4)	`Polynomial.binomExpansion`	[Tay15]
Congruences; $\mathbb{F}_p[X]$ a domain (Lemma 5)	`ZMod`, `Polynomial.map`, `Nat.Prime`	[Gau01], §I
Algebraic $\Leftrightarrow$ integral	`isAlgebraic_iff_isIntegral`, `IsIntegral.of_mem_of_fg`	[Ded71]
Degree arithmetic; associates, units, prime vs. irreducible	`natDegree_mul`, `natDegree_comp`, `natDegree_le_of_dvd`, `Associated.pow_pow`, `isUnit_of_dvd_unit`, `Irreducible.prime`	classical
`Algebra`, `AlgEquiv`, `RingHom` scaffolding	`AlgEquiv.ofAlgHom`, `aeval_map_algebraMap`, `MulEquiv.irreducible_iff`	no mathematical content

Three entries deserve a word. The attribution of localization to Grell [Gre27] is the standard one, though rings of fractions were in use earlier and reached their general form later. Mathlib states the notion by its universal property, and only the denominator-clearing corollary is used here. The row for degree arithmetic is deliberately blank: these are the facts every algebra course proves and no paper claims. And the last row is scaffolding — statements about Lean's algebraic hierarchy rather than about mathematics.

Two gaps in Mathlib are worth recording in the same spirit, since they are the places where the development could not import and had to build. Neither changes any statement, but they are gaps of different kinds.

`Polynomial.binomExpansion` is stated for `eval` rather than for `comp`, so Lemma 4 is obtained by transporting it along $C : \mathbb{Z} \to \mathbb{Z}[X]$: composition *is* evaluation once the coefficients are pushed up into $\mathbb{Z}[X]$. That is eight lines and no mathematics.

The dilation is not notation. Mathlib has the translation $X \mapsto X + c$ as an algebra equivalence but not $X \mapsto cX$, and the reason is that the latter is an automorphism only when c is invertible. Over $\mathbb{Z}$ the map $p \mapsto \tilde{p}$ is injective but not surjective, and it does not preserve irreducibility: for $h = X + 3$ with $h_0 = 3$ and $\rho = 1$, so $M = 3$,

$$\tilde{h} = 3X + 3 = 3(X + 1)$$

is reducible in $\mathbb{Z}[X]$. This is systematic rather than accidental. Since $h_0 \mid M$, every coefficient $h_j M^j$ with $j \geq 1$ is divisible by h_0, and the constant coefficient is h_0 itself, so h_0 always divides the content of $\widetilde{h}$. Dividing it out is at once what makes H integral in Lemma 1 and what removes the obstruction to irreducibility, and $H(0) = 1$ then makes H primitive. So Lemma 7 has to pass through $\mathbb{Q}$, where the dilation is an automorphism with inverse the dilation by M^{-1}, and return by Gauss — and it would have to do so whether or not Mathlib supplied the equivalence. The gap cost twenty-five lines; it cost the argument nothing.

References

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